Comparative Learning Strategies for Training Belief Rule Base Systems: A Case Study of E-Government Assessment*

Tanjim Mahmud[1], Puja Dey[2], Md Akash Rahman[3], Mohammad Shahadat Hossain[2,4], and Karl Andersson[4]

[1] Rangamati Science and Technology University, Rangamati 4500, Bangladesh
tanjim_cse@yahoo.com
[2] University of Chittagong, Chittagong -4331, Bangladesh
pujadeycu1@gmail.com hossain_ms@cu.ac.bd
[3] Port City International University, Chattogram, Bangladesh
akashrahman.cse@gmail.com
[4] Luleå University of Technology, 97187 Luleå, Sweden
karl.andersson@ltu.se

Abstract

E-Government short for electronic government, also known as e-gov, digital government consists of the digital interactions between a government and citizens, government and businesses/Commerce, government and employees, and also between government and governments /agencies. E-government is no longer just an option but a necessity for countries aiming for better governance. The e-government performance assessment is important for the development of e-government. As conventional evaluation models for e-government performance are generally too subjective and inaccurate, this study proposed belief Rule-base Inference Methodology using the Evidential Reasoning approach (RIMER) method to evaluating the e-government performance.The aim of the research is to evaluate optimal performance of E-government. The method will generate result an accurate system for E-government assessment.In the research, we employed belief Rule-base Inference Methodology using the Evidential Reasoning approach (RIMER) for design and development of belief rule-based expert system. In the system, belief rule base (BRB) is used to model uncertain E-Government domain knowledge, the evidential reasoning (ER) approach is employed to build inference engine, a BRB training module is developed for learning the BRB through accumulated a lot of cases. We use a set of simulated data and to validate the developed belief rule-based expert system prototype. The results show that the prototype can provide reliable recommendations and the performance of the system can be improved significantly after training BRB using accumulated different cases.The expert system is designed and developed by using Visual Basic, MS SQL Server and Mat lab. It can show that the developed Expert System can generate appropriate results taking uncertain information.The achievement of the system is that it has assessed optimal performance of E-government. The developed system handles uncertainty in an efficient manner and generates real system output.

1 Introduction

Belief Rule Based Expert Systems (BRBES) utilize the Belief Rule Base (BRB) for knowledge representation and Evidential Reasoning (ER) for inference [35] [17, 23]. BRB extends

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traditional IF–THEN rules by incorporating belief degrees in the consequent, enabling the representation of expert knowledge under uncertainty. ER, developed from Dempster–Shafer theory, handles ignorance [25], while BRB integrates fuzzy logic to address vagueness and imprecision. Their combined inference methodology, RIMER, can simultaneously manage vagueness, imprecision, ignorance, and incompleteness while processing both qualitative and quantitative data [38] [16, 21] [35]. This integrated capability distinguishes BRBES from Bayesian Belief Networks, fuzzy logic, and D–S theory when used independently.

BRBES has been successfully applied in domains characterized by uncertainty, including health informatics, flood prediction, affective computing, and E-Government evaluation [33]. In E-Government assessment, twenty-one variables with different measurement scales require effective handling of uncertainty, as demonstrated by Hossain et al. [18, 31] [13, 19] [14] [10, 12]. However, optimizing BRBES through effective training remains a challenge, requiring appropriate objective functions, parameter selection, and knowledge base updating strategies. Therefore, this paper proposes a learning framework to improve BRBES performance using E-Government assessment as a case study.

Although BRB and RIMER methodologies have previously been applied to E-Government assessment, this study makes several distinct contributions:

- A hierarchical BRB framework consisting of 21 E-Government assessment variables organized into Determinants, Characteristics, and Results categories.
- An integrated learning framework to investigate the impact of training different BRB parameters, including rule weights, attribute weights, belief degrees, and utility scores.
- A comparative evaluation of five training configurations to identify the most effective parameter subset for reducing prediction error while mitigating overfitting.
- Validation that BRB training improves the predictive capability of E-Government assessment systems under uncertain and incomplete information.

Unlike the generic BRB optimization framework proposed by Yang et al. [34], this study comparatively evaluates different parameter-training strategies for E-Government assessment to identify their practical effectiveness.

The remainder of this paper is organized as follows. Section 2 reviews related work, Section 3 presents the proposed methodology, Section 4 describes the BRBES prototype, Section 5 discusses the experimental results, and Section 6 concludes the paper.

2 Existing E-Government Evaluation Approaches

Evaluating government performance in ICT project implementation from the citizens' perspective requires an appropriate research approach. Governments worldwide invest substantial financial resources, often exceeding 1% of GDP, in developing e-government services, making effective evaluation essential to justify these investments [26] [1]. Traditional evaluation methods, such as return on investment (ROI), cost-benefit analysis, payback period, and present worth, are widely used but have limitations in assessing e-government projects [1, 3] [6]. These approaches mainly focus on tangible financial outcomes while overlooking human, organizational, and intangible benefits [22] [8]. To overcome these limitations, several citizen-centric evaluation models have been proposed. Wang et al. developed a framework for evaluating e-government services that explains the success or failure of government portals in meeting citizens' information needs [11, 27] [1]. Similarly, Eschenfelder et al. introduced a socio-technical toolkit that

evaluates the openness of e-government portals by considering information characteristics, social and political contexts, stakeholder value, and citizen participation [5]. Carter and Bélanger applied the Technology Acceptance Model (TAM) and Diffusion of Innovation (DOI) theory to identify factors influencing citizen adoption of e-government services, including perceived usefulness, compatibility, ease of use, image, and trust in government and the Internet [4]. Fuzzy set theory, introduced by Zadeh in 1965, provides an effective framework for handling vague, ambiguous, and imprecise information by assigning degrees of membership between 0 and 1 [20]. Unlike classical set theory, it supports gradual transitions between sets and enables reasoning under uncertainty, making it suitable for complex decision-making problems. Its ability to represent possibility, vagueness, and nonlinear integration has led to applications in engineering, information processing, and social science research [2].

3 Methods

A belief degree is associated with each possible consequent of a rule in a Belief rule-base(BRB) expert system. Belief Rule base knowledge representation schema has the ability to handle uncertainty. In belief rule base expert system, the knowledge base of the expert system is developed using Evidential Reasoning approach which is the replacement of inference engine in the knowledge base [28] [33] [29]. In this approach, a rule base is designed with belief degrees embedded in all possible consequents of a rule. Complicated and continuous causal relationships between different factors can be captured by belief rule-base expert system which is not possible for traditional IF-THEN rules.

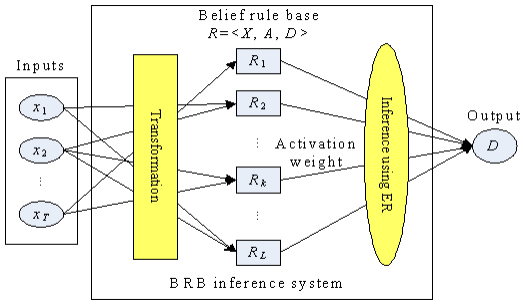


Figure 1: Single-layer BRB inference architecture with RIMER Methodology

3.1 BRB for Representing E-government Assessment Domain Knowledge

Belief Rule Bases (BRBs) extend traditional rule bases by incorporating a belief structure into the consequent part of the rules. In a belief rule, each antecedent attribute takes referential values, while each consequent is associated with corresponding belief degrees [33]. The knowledge representation also embeds rule weights, attribute weights, and belief degrees.

The kth rule R_k in a traditional IF–THEN rule base is expressed as

$$\text{If } A_{1k} \wedge A_{2k} \wedge \cdots \wedge A_{T_k k}, \text{ then } D_k \tag{1}$$

where

A_{ik} : Referential value of the ith antecedent attribute in the kth rule,
T_k : Number of antecedent attributes in the kth rule,
D_k : Consequent of the kth rule.

By incorporating belief degrees, rule weights, and attribute weights, the traditional rule is extended into a belief rule as

$$\text{If } A_{1k} \wedge A_{2k} \wedge \cdots \wedge A_{T_k k}, \text{ then } \{(D_1, \beta_{1k}), (D_2, \beta_{2k}), \ldots, (D_N, \beta_{Nk})\} \tag{2}$$

with rule weight θ_k and attribute weights $\delta_{k1}, \delta_{k2}, \ldots, \delta_{kT_k}$, where

β_{ik} : Belief degree associated with the ith consequent,
δ_{ki} : Relative importance of the ith antecedent attribute in the kth rule,
θ_k : Relative importance (weight) of the kth rule,
L : Total number of belief rules in the rule base,
T_k : Number of antecedent attributes in the kth belief rule.

3.2 BRB Inference System

The Evidential Reasoning (ER) approach is widely used for multi-attribute decision analysis (MADA) under uncertain quantitative and qualitative information [15, 24]. In a Belief Rule Base (BRB), each antecedent A_k $(k = 1, \ldots, L)$ is associated with a weight W_k and linked to consequents D_i $(i = 1, \ldots, N)$ through belief degrees β_{ik}. The belief rule structure is summarized in Table 15.

For an input vector $U = (U_i, i = 1, \ldots, T)$ with belief degrees $\epsilon = (\epsilon_i, i = 1, \ldots, T)$, each input is transformed into a distributed belief structure:

$$S(U_i, \epsilon_i) = \{(A_{ij}, x_{ij}); j = 1, \ldots, J_i\}.$$

The activation weight of the kth rule is computed as:

$$W_k = \frac{\theta_k \alpha_k}{\sum_{i=1}^{L} \alpha_i \theta_i},$$

where θ_k is the rule weight and α_k is the matching degree. Expert belief degrees $\bar{\beta}_{ik}$ may be complete or incomplete and are updated during inference as:

$$\beta_{ik} = \bar{\beta}_{ik} \frac{\sum_{t=1}^{T_k} \tau(t,k) \sum_{j=1}^{J_t} \alpha_{tj}}{\sum_{t=1}^{T_k} \tau(t,k)}.$$

The ER algorithm aggregates all activated rules to produce the output belief distribution:

$$O(U) = \{(D_j, \beta_j), j = 1, \ldots, N\}.$$

3.2.1 Input Transformation

In this research, the referential set {Excellent, Very Good, Good, Satisfactory, Poor} is used for all E-government assessment factors. Each input is transformed into a distributed belief structure:

$$S(A_i^*) = \{(h_{3i}, \gamma_{3i}), (h_{2i}, \gamma_{2i})\},$$

where

$$\gamma_{2i} = \frac{h_{3i} - A_i}{h_{3i} - h_{2i}}, \quad \gamma_{3i} = 1 - \gamma_{2i}.$$

The transformation follows a rule-based mapping approach [32].

3.2.2 Qualitative input transformation

Qualitative inputs are mapped to the referential values {Excellent, Very Good, Good, Satisfactory, Poor} using expert-defined rules. Typically, categories such as *Excellent* and *Good* are assigned higher belief degrees based on domain knowledge, while less relevant categories receive lower or zero values. These mappings are determined by expert judgment.

3.2.3 Quantitative input transformation

Each quantitative attribute is associated with lower and upper threshold values defined by domain experts. If an input is below the lower bound, it is mapped to 100% Good; if above the upper bound, it is mapped to 100% Very Good; otherwise, a linear transformation is applied using the equation above.

3.2.4 Rules activation weights calculation

After assigning antecedent referential values, the next step is to compute rule activation weights. First, the normalized attribute weights are obtained as:

$$\overline{\delta}_{ki} = \frac{\delta_{ki}}{\max\{\delta_{ki}\}}, \quad i = 1, \ldots, T_k$$

The combined matching degree of each rule antecedent is calculated as:

$$\delta_k = \prod_{i=1}^{T_k} (\alpha_{ik}) \delta_{ki}$$

The activation weight of the kth rule is then given by:

$$W_k = \frac{\theta_k \alpha_k}{\sum_{i=1}^{L} \alpha_i \theta_i}$$

These computations determine the degree to which each rule is activated based on input matching.

3.2.5 Belief-degree Update

The belief degree β_{ik} is initially assigned by experts during BRB construction. A rule is considered complete if $\sum \beta_{ik} = 1$, and incomplete otherwise. If $\sum \beta_{ik} = 0$, the rule output is completely unknown. Incompleteness may arise from limited expert knowledge or missing input information. In such cases, both antecedent and consequent uncertainty must be retained in the inference process [**?**]. When input data is incomplete, the resulting inference also becomes incomplete [33]. Therefore, belief degrees must be updated according to the actual input to reflect partial or missing information during inference.

3.2.6 Rule aggregation via ER

The Evidential Reasoning (ER) approach aggregates all activated rules for multi-attribute decision analysis under uncertainty [9] [36] [30]. Unlike conventional MADA methods, ER uses a belief distribution framework instead of a crisp decision matrix. In this model, each rule acts as a basic attribute, and its activation weight represents its importance, while consequents correspond to evaluation grades.

3.2.7 Output of the BRB System

The output of a BRBES is a belief distribution rather than a single numerical value. It can be converted into a crisp score using a utility function assigned to each consequent grade [33]:

$$H(A^*) = \sum_{j=1}^{N} u(C_j) B_j.$$

3.3 Optimal Learning Methods for Training Belief Rule Bases in RIMER

In this research, the initial BRB is constructed using expert knowledge and validated E-government rules. Since domain knowledge evolves slowly, BRB parameters are optimized through offline learning using historical E-government datasets. Several optimization-based training methods have been proposed in [34]. When outputs are numerical, training is formulated as a nonlinear optimization problem that minimizes the mean squared error (MSE) between predicted and observed values.

The training parameters include rule weights (θ_k), attribute weights (δ_j), and belief degrees (β_{jk}). Given a training set of M samples $(\hat{x}_m, \hat{y}_m)$, where $m = 1, \ldots, M$, the model output y_m is generated by the inference engine, and $\hat{y}_m$ is the observed output. The objective function is:

$$\xi(P) = \frac{1}{M} \sum_{m=1}^{M} (y_m - \hat{y}_m)^2.$$

Inference is performed using the analytical ER algorithm [28], where the aggregated belief degrees for consequents D_j $(j = 1, \ldots, N)$ are computed as:

$$\beta = \frac{\mu \left[\prod_{k=1}^{L} \left(\omega_k \beta_{jk} + (1 - \omega_k) \sum_{j=1}^{N} \beta_{jk} \right) - \prod_{k=1}^{L} \left(1 - \omega_k \sum_{j=1}^{N} \beta_{jk} \right) \right]}{1 - \mu \left[\prod_{k=1}^{L} (1 - \omega_k) \right]},$$

$$\mu = \left[\sum_{j=1}^{N} \prod_{k=1}^{L} \left(\omega_k \beta_{jk} + (1 - \omega_k) \sum_{j=1}^{N} \beta_{jk} \right) - (N-1) \prod_{k=1}^{L} \left(1 - \omega_k \sum_{j=1}^{N} \beta_{jk} \right) \right]^{-1}.$$

Here, β_{jk} is the belief degree of the jth consequent in the kth rule, and ω_k is the activation weight of the kth rule. Since ER produces a belief distribution (D_j, β_j) rather than a single value, the final output is obtained using a utility function [37]:

$$y = \sum_{j=1}^{N} u(D_j)\beta_j.$$

Optimization Constraints

The optimization is subject to the following constraints:

$$0 \leq \theta_k \leq 1 \tag{1}$$

$$0 \leq \delta_j \leq 1 \tag{2}$$

$$\sum_{j=1}^{N} \beta_{jk} \leq 1 \tag{3}$$

where θ_k is the rule weight, δ_j is the attribute weight, and β_{jk} is the belief degree.

The optimization terminates when the reduction in MSE between successive iterations falls below a threshold or when the maximum number of iterations is reached. These constraints ensure consistency of BRB parameters and prevent infeasible solutions during training.

4 Design and Development of Belief Rule-based Expert System Prototype

In this research, a belief rule-base expert system (BRBES) is developed for E-government assessment using multiple evaluation factors. The BRB framework extends traditional rule-based systems by incorporating belief structures, enabling effective handling of uncertainty. Inference is performed using the Evidential Reasoning (ER) approach, which supports the fusion of both qualitative and quantitative uncertain information in E-government assessment.

4.1 System Architecture

The system architecture defines the structure of inputs, processing, and outputs. The proposed BRBES follows a three-layer design: (i) interface layer for collecting input referential values, (ii) application layer containing the inference engine, and (iii) data processing layer where the initial BRB knowledge base is stored.

Figure 2 illustrates the proposed three-layer architecture.

Figure 3 presents the inference engine structure used in the application layer.

The Recursive ER algorithm used in the inference engine is summarized as follows:

1. Step 1: Start

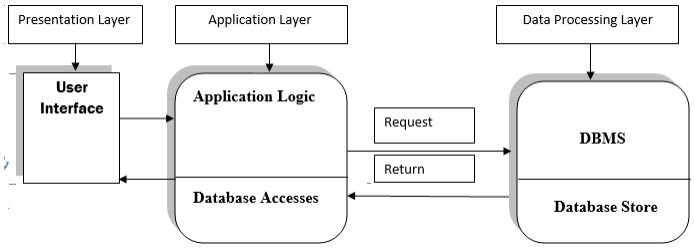


Figure 2: Three-Layer Architecture

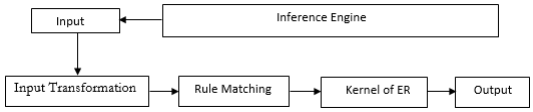


Figure 3: Inference-engine

2. Step 2: Read rule activation weights and updated consequent belief degrees from inputs.
3. Step 3: Compute belief degrees and unassigned belief from the first activated rule.
4. Step 4: Combine current rule with previously aggregated rules.
5. Step 5: Repeat until all rules are processed.
6. Step 6: Normalize final belief degrees.
7. Step 7: End.

4.2 Knowledge Base

The knowledge base is constructed using a BRB framework as shown in Figure 4. To reduce complexity, nine "Characteristics" variables are grouped into three subgroups: C1 (User Environment), C2 (Resource Management), and C3 (Authentication Protocol). Similarly, seven "Result" variables are grouped into R1 (Result Analysis) and R2 (Result Specification).

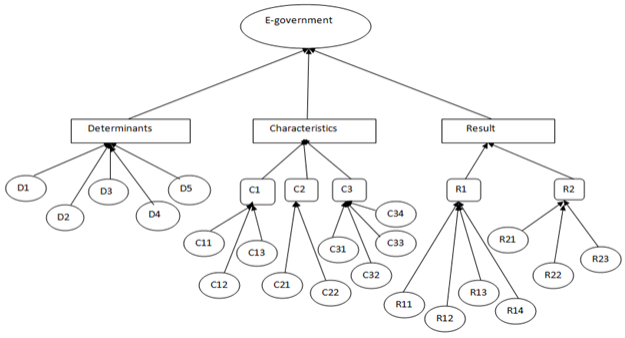


Figure 4: Hierarchical Relationship among E-Government Assessment Factors

Knowledge Acquisition and Rule Construction

The initial BRB is constructed using expert knowledge and literature-based E-government evaluation criteria. Five referential values are used for each antecedent: Excellent, Very Good, Good, Satisfactory, and Poor. With five antecedents, the total number of rules is:

$$5^5 = 3125$$

Initial belief distributions are assigned based on expert judgment and later refined using training data.

The determinants considered are: D1: Quality of information and data availability D2: Technological infrastructure and compatibility D3: Organizational and management factors D4: Legal and institutional framework D5: Potential demand

Characteristics include: C1: User environment (Interaction, Integration, Personalization) C2: Resource management (Information quality, Services) C3: Authentication protocol (Security, Accessibility, Usability, Privacy)

Results include: R1: Result analysis (Service quality, Efficiency, Effectiveness, Transparency) R2: Result specification (System usage, Citizen participation, Regulatory change)

4.3 Trained Module Design Method

Overfitting occurs when a model performs better on training data but worse on unseen test data, especially when the parameter space is overly complex (Mitchell, 1997). In BRB training, overfitting may arise when too many parameters are optimized simultaneously.

To address this issue, five training configurations are designed:

R1: rule weights, attribute weights, belief degrees, and utility scores **R2:** rule weights, attribute weights, and belief degrees **R3:** attribute weights and belief degrees **R4:** rule weights and belief degrees **R5:** belief degrees only

The training and validation procedure is as follows:

1. Select training and testing samples from 1000 simulated cases.
2. Train BRB using the five configurations with identical initial parameters.
3. Evaluate performance on training and test datasets.
4. Analyze results using ROC curves (Chapter 7).
5. Draw conclusions on optimal training configuration.

4.4 Graphical user interface of Belief Rule Based Expert System

The component implemented in the prototype which concerns the interaction between users and the system is user interfaces. There are many software available for developing user interfaces. Among them Microsoft Visual Basic 6.0 software is used at front end to develop this project. Because it is easy to implement our desired interfaces using this software. This software gives us the facility of making a link between SQL Server and the user interfaces developed by it. Through this language we can implement any declarative query such as search and show which is most important for my software. Moreover Visual Basic 6.0 provides us the facilities of importing different images and pictures. We can making any Labels, buttons, textboxes, frames, tabs etc without writing any code like other languages such as Java, C etc. Through this data reports of different tables of database can be easily generated and printed. User interfaces is developed by MS Visual Basic 6.0

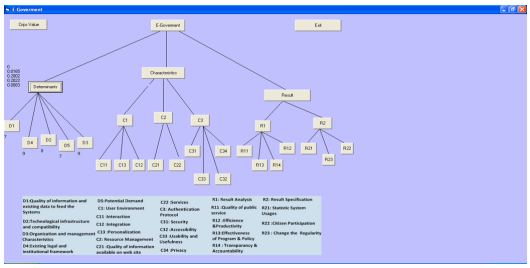


Figure 5: BRBES graphical user interface

5 Results and Discussion

The developed BRB-based Expert System for E-Government performance assessment can be applied to evaluate various aspects of E-Government initiatives. The system employs the RIMER methodology to effectively manage different forms of uncertainty, including vagueness, incompleteness, and imprecision. In the proposed framework, a Belief Rule Base (BRB) is first constructed and subsequently combined with Evidential Reasoning (ER) for inference generation. This approach addresses several limitations of traditional evaluation methods and provides a more reliable mechanism for E-Government performance assessment.

Dataset Description

The experimental evaluation was conducted using 20 representative E-Government assessment cases. Each case consists of five input variables (D_1–D_5) and one observed output value representing the E-Government performance level.

The dataset was constructed based on expert knowledge and previous E-Government assessment studies. Due to the limited availability of real-world assessment data, the current study utilizes a relatively small dataset to demonstrate the effectiveness of the proposed BRB training framework.

For training and testing purposes, the first nine cases were used for model training, while the remaining nine cases were used for testing and validation. The complete dataset used in this study is summarized in Table 1.

The given input values and observed outputs are represented as numerical values. The output data are subsequently transformed into belief structures through the input transformation process [32,33]. The resulting belief distributions are used during BRB inference and parameter training.

Table 1: Training Data Set (Input and Observed Output Values)

Set	Given Input					Observed Output(D)					
	D1	D2	D3	D4	D5	Numerical	Judgment with belief				
1	1	0.88	0.88	0.6	0.3	1	0	0.2667	0.7333	0	0
2	0.66	0.56	0.77	0.88	0.55	0.9	0	0.1333	0.8667	0	0
3	0.4	0.7	0.78	0.44	0.55	0.6	0	0	1	0	0
4	1	0.88	0.88	0.6	0.3	0.4	0	0.4	0.6	0	0
-	-	-	-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-	-	-	-
20	0.4	0.7	0.78	0.44	0.55	0.88	0	0	0.3333	0.6667	0

Table 2: Initial BRB and Weights

Sl No	Rule Weight	Excellent	Very Good	Good	Satisfactory	Poor
0	1	1	1	0	0	0
1	1	0.6	0.4	0	0	0
2	1	0.4667	0.5333	0	0	0
3	1	0.3333	0.6667	0	0	0
4	1	0.2	0.8	0	0	0
-	-	-	-	-	-	-
-	-	-	-	-	-	-
3124	1	0	0.6667	0.3333	0	0

5.1 Trained BRB and Rule Weights

We tried five different sets of training parameters for the training module. In the following discussion, we will use R1, R2, R3, R4, and R5 to represent the BRB training with different parameters, where R1: training with rule weights, antecedent attribute weights, consequent belief degrees and severity scores of the four different risk levels R2: training with rule weights, antecedent attribute weights, and consequent belief degrees R3: training with antecedent attributes weights and consequent belief degrees R4: training with rule weights and consequent belief degrees R5: training with belief degrees

Table 3: Training with Rule Weight, Antecedent Attribute, Consequent Belief Degree and Utility Score

Sl No	Rule Weight	Attribute Weight	Excellent	Very Good	Good	Satisfactory	Poor	Utility Score
0	0.0667	0	0.4667	0.5333	0	0	0	0
1	0.2	0.4	0.0667	0.9333	0	0	0	0.4
2	0.3333	0.5333	0	0.9333	0.0667	0	0	0.5333
3	0.4667	0.6667	0	0.8	0.2	0	0	0.6667
4	0.0667	0.8	0	0.8	0.3333	0	0	0.8
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
3124	0.8667	0.6667	0	0.3333	0.3333	0	0	0.6667

5.2 Comparison between Observed and Trained Result

This subsection shows the comparisons between the observed output and the trained output of our expert system.

5.3 Observed and Simulated Values of the Output

For the validation of trained approaches, the obtainable dataset is separated into two sets : training and testing. The training dataset is utilised to train parameters of the system. Consequently, this trained BRB system is applied to create the output which would be the input of test data. We use the first 9 sets of data from the Table 1 for training purpose. Then

Table 4: Training with Rule Weights, Antecedent Attribute Weights, And Consequent Belief Degrees

Sl No	Rule Weight	Attribute Weight	Excellent	Very Good	Good	Satisfactory	Poor
0	0.4667	0	0	0	0.4667	0.5333	0
1	0.3333	0.1333	0	0.1333	0.8667	0	0
2	0.2	0	0	0	0.7333	0.2667	0
3	0.6	0	0	0	0.6	0.4	0
4	0.7333	0	0	0	0.4667	0.5333	0
-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-
3124	0.55	0	0	0	1	0	0

Table 5: Training with antecedent attributes weights and consequent belief degrees

Sl No	Attribute Weight	Excellent	Very Good	Good	Satisfactory	Poor
0	0	0	0	0.8667	0.1333	0
1	0	0	0	0.4667	0.5333	0
2	0	0	0	0.3333	0.6667	0
3	0	0	0	0.2	0.8	0
4	0	0	0	0.0667	0.9333	0
-	-	-	-	-	-	-
-	-	-	-	-	-	-
3124	0	0	0	1	0	0

Table 6: Training with Rule Weights and Consequent Belief Degrees

Sl No	Rule Weight	Excellent	Very Good	Good	Satisfactory	Poor
0	0.9333	0	0.9333	0.0667	0	0
1	0.5333	0	0.5333	0.4667	0	0
2	0.4	0	0.4	0.6	0.6667	0
3	0.2667	0	0.2667	0.7333	0	0
4	0.1333	0	0.1333	0.8667	0	0
-	-	-	-	-	-	-
-	-	-	-	-	-	-
3124	0.2667	0	0.2667	0.7333	0	0

the second 9 sets of data are utilised for testing. Table 13 shows the observed and simulated outputs. Figure 7 and Figure 8 are the graphical representation of the observed output and the simulated output.

5.4 Receiver operating Characteristics (ROC) Curve

Assessing the predictive accuracy is essential for implementing and comparing models, technologies , or algorithms that generate predictions. The ROC curve offer a magnificent and visually appealing approach to summarize the prediction accuracy [7]. So, in our research, we utilized ROC curves for measuring the accuracy of the e-government evaluation produced by

Table 7: Training with Belief Degrees

Sl No	Excellent	Very Good	Good	Satisfactory	Poor
0	0	0	0	0.5333	0.4667
1	0	0	0.2	0.8	0
2	0	0	0	0.8	0.2
3	0	0	0	0.6667	0.3333
4	0	0	0	0.5333	0.4667
-	-	-	-	-	-
-	-	-	-	-	-
3124	0	0	0	0.9333	0.0667

Table 8: Training with Rule Weight, Antecedent Attribute, Consequent Belief Degree and Utility Score(R1)

Number of Training	1	2	3	4	5	6	7	8	9
Observed Output	1	0.9	0.6	0.4	0.6	0.4	0.7	0.8	0.4
Trained	0.99	0.8	0.57	0.35	0.58	0.5	0.75	0.77	0.34
Number of Training	10	11	12	13	14	15	16	17	18
Observed Output	0.77	0.4	0.3	0.33	0.4	0.56	0.7	0.6	0.88
Trained	0.76	0.44	0.28	0.29	0.35	0.51	0.66	0.55	0.78

Table 9: Training with Rule Weights, Antecedent Attribute Weights, And Consequent Belief Degrees (R2)

Number of Training	1	2	3	4	5	6	7	8	9
Observed Output	1	0.9	0.6	0.4	0.6	0.4	0.7	0.8	0.4
Trained	0.88	0.8	0.44	0.45	0.63	0.34	0.67	0.77	0.44
Number of Training	10	11	12	13	14	15	16	17	18
Observed Output	0.77	0.4	0.3	0.33	0.4	0.56	0.7	0.6	0.88
Trained	0.67	0.44	0.28	0.24	0.35	0.54	0.78	0.53	0.76

the Belief Rule-based Expert System Prototype.

For ROC analysis, the continuous BRB output was converted into a binary decision variable using an expert-defined threshold. Outputs above the threshold were considered satisfactory E-Government performance, whereas outputs below the threshold were considered unsatisfactory performance.

The ROC curve therefore evaluates the ability of the trained BRB models to discriminate between satisfactory and unsatisfactory assessment outcomes.

Table 10: Training with antecedent attributes weights and consequent belief degrees (R3)

Number of Training	1	2	3	4	5	6	7	8	9
Observed Output	1	0.9	0.6	0.4	0.6	0.4	0.7	0.8	0.4
Trained	0.77	0.67	0.67	0.45	0.63	0.34	0.67	0.88	0.41
Number of Training	10	11	12	13	14	15	16	17	18
Observed Output	0.77	0.4	0.3	0.33	0.4	0.56	0.7	0.6	0.88
Trained	0.67	0.44	0.28	0.24	0.41	0.5	0.73	0.61	0.76

Table 11: Training with Rule Weights and Consequent Belief Degrees (R4)

Number of Training	1	2	3	4	5	6	7	8	9
Observed Output	1	0.9	0.6	0.4	0.6	0.4	0.7	0.8	0.4
Trained	0.67	0.44	0.31	0.32	0.41	0.5	0.73	0.61	0.76
Number of Training	10	11	12	13	14	15	16	17	18
Observed Output	0.77	0.4	0.3	0.33	0.4	0.56	0.7	0.6	0.88
Trained	0.6	0.44	0.31	0.54	0.41	0.6	0.73	0.61	0.76

Table 12: Training with Belief Degrees (R5)

Number of Training	1	2	3	4	5	6	7	8	9
Observed Output	1	0.9	0.6	0.4	0.6	0.4	0.7	0.8	0.4
Trained	0.99	0.89	0.62	0.42	0.64	0.43	0.73	0.81	0.41
Number of Training	10	11	12	13	14	15	16	17	18
Observed Output	0.77	0.4	0.3	0.33	0.4	0.56	0.7	0.6	0.88
Trained	0.67	0.44	0.31	0.32	0.41	0.5	0.73	0.61	0.76

Table 13: Observed and Simulated Values of the Output

Test Data Set	10	11	12	13	14	15	16	17	18
Observed Output	0.77	0.4	0.3	0.33	0.4	0.56	0.7	0.6	0.88
Simulated Output	0.76	0.44	0.31	0.33	0.42	0.57	0.77	0.64	0.89

Following table 14 describes the ROC Curves.

The ROC analysis demonstrates that the trained BRB configurations achieved improved

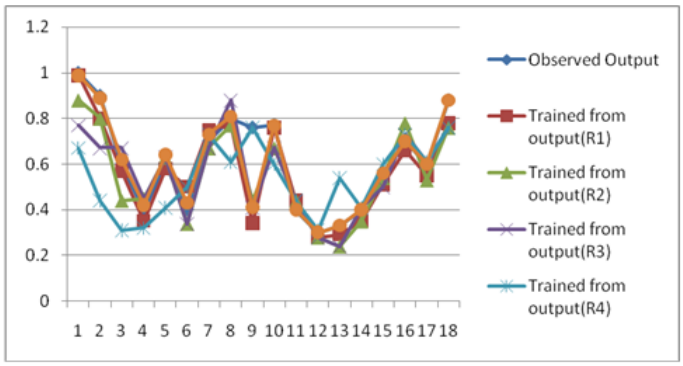
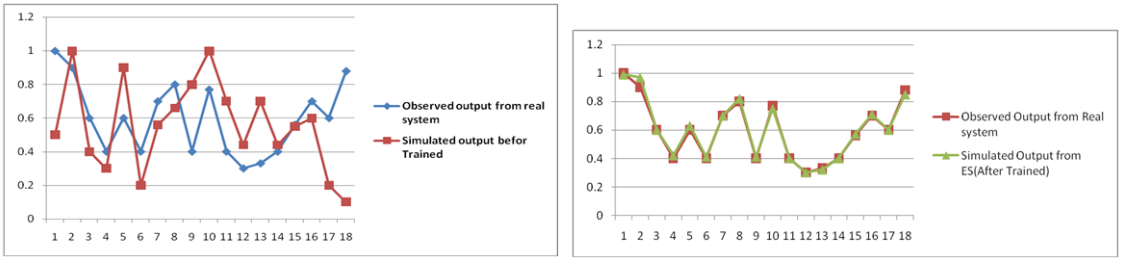


Figure 6: R1, R2, R3, R4, R5 vs Observed Output



(a) Result (Before Trained) (b) Result (After Trained)

Figure 7: Comparison of results before and after training

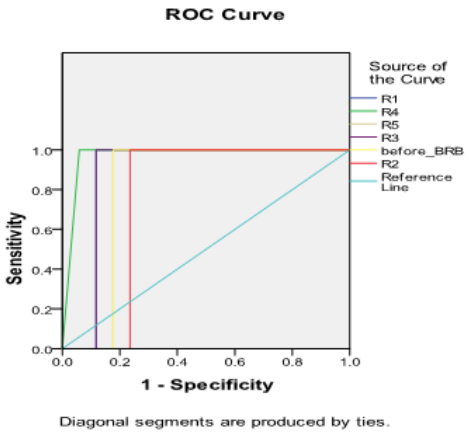


Figure 8: Result (ROC Curve)

discriminative performance compared with the baseline model. The observed AUC values indicate the effectiveness of the proposed learning framework in enhancing the predictive capability of the BRB-based expert system.

The results further suggest that appropriate training of BRB parameters can contribute

Table 14: Area Under the Curve

Test Result Variable(s)	Area	Std. Error	Asymptotic Sig.	Asymptotic 95% Confidence Interval	
				Lower Bound	Upper Bound
before-BRB	.824	.092	.289	.637	1.000
R1	.882	.078	.210	.000	1.000
R2	.765	.103	.386	.563	.966
R3	.882	.078	.210	.000	1.000
R4	.971	.044	.123	.000	1.000
R5	.824	.092	.289	.637	1.000

to more accurate E-Government performance assessment under uncertain environments. In addition to ROC and AUC analysis, future research will explore complementary evaluation metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Precision, Recall, and other validation measures to provide a broader assessment of system performance across different application scenarios.

Table 15: Comparison between Real System Output and Expert System Output

Observed Output(Real System Output)	Simulated Output After Trained (Expert System Output)
Excellent (0%)	Excellent (0.5%)
Very Good (0%)	Very Good (0%)
Good (73%)	Good (72%)
Satisfactory (27%)	Satisfactory (27.5%)
Poor (0%)	Poor (0%)

6 Conclusion and Future Work

6.1 Conclusion

Evaluating E-government initiatives is essential for determining the success or failure of projects and identifying areas that require improvement. In Bangladesh, most ministries have begun providing basic E-government services through their websites. However, the full benefits of these services cannot be realized until more advanced transactional features are implemented. Furthermore, challenges such as limited awareness among older citizens, bureaucratic practices, and socio-cultural barriers must also be addressed.

This study highlights the growing importance of E-government in enhancing convenience and accessibility for citizens. The younger generation, with higher internet literacy, shows a strong willingness to use online government services due to their efficiency in saving time and cost. Nevertheless, improving the quality and usability of E-government websites remains crucial, as poorly designed systems may discourage citizens from using these services.

In this research, a Belief Rule Base (BRB)-based Expert System (ES) was developed to evaluate E-government projects using data collected from three districts of Bangladesh across 21 variables. The proposed system enables continuous assessment of E-government initiatives and helps identify variables that negatively affect project performance, thereby supporting better policy formulation. Moreover, the BRB-based ES can be applied at different administrative

levels, such as divisional, district, upazila, and union levels, and may also be extended to evaluate ICT projects in sectors such as education, agriculture, and healthcare. The proposed framework can potentially be adapted for evaluating E-government systems in other countries.

6.2 Limitations

Several limitations should be acknowledged in the present study. The validation dataset is relatively small and partially based on simulated assessment cases. In addition, the implementation platform utilizes legacy technologies including Visual Basic 6.0 and Microsoft SQL Server, which may limit scalability and reproducibility in large-scale deployment environments. Furthermore, the current study focuses on district-level E-Government assessment in Bangladesh; therefore, the findings may require further validation before being generalized to different countries and administrative contexts. Future research will address these limitations through the use of larger real-world datasets, modern software implementation platforms, and cross-country validation studies.

6.3 Future Work

Future research will also focus on improving the system by expanding the knowledge base with a larger set of rules and enhancing the BRB training process for improved validation and accuracy. Additionally, an online BRB-based expert system will be developed to enable more robust, scalable, and real-time evaluation of E-government projects.

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